Electron diffusion at operating pressure

October 29, 2026

Operating pressure: 15 bar. Xe-He helium fraction: 15% (user design inputs). Petit prescription: sigma_T,L [mm] = d_T,L [mm sqrt(bar/cm)] sqrt(L_cm/P_bar). The operating-pressure coefficients below multiply sqrt(L_cm). Pure-Xe reference coefficients are 3.5/0.9; Xe-He coefficients are 1.6/0.75. Widths use Float32; maximum absolute difference from Float64 evaluation is 5.81e-07 mm. The electron formula uses the supplied coefficients and pressure; it has no explicit electric-field argument.

Gas	$d_T(15\,\mathrm{bar})$ [mm/$\sqrt{\mathrm{cm}}$]	$d_L(15\,\mathrm{bar})$ [mm/$\sqrt{\mathrm{cm}}$]
Pure Xe	0.903696	0.232379
Xe + 15% He	0.413118	0.193649

Gas	Drift [cm]	σ_T [mm]	σ_L [mm]
Pure Xe	10	2.8577	0.7348
Pure Xe	100	9.0370	2.3238
Pure Xe	150	11.0680	2.8460
Xe + 15% He	10	1.3064	0.6124
Xe + 15% He	100	4.1312	1.9365
Xe + 15% He	150	5.0596	2.3717